

# The German Annual Grid Carbon Factor Is Hiding an 8.7× Problem

*Hourly Carbon-Accounting Check on Germany's 2025 Electricity Grid*

## 1. Why I Looked

A few weeks ago I posted about Germany's 2024 grid-congestion cost: €2.9 billion spent managing a system that curtailed 9.4 TWh of renewable generation it had already made. The post was about a coordination failure. But the question that stayed with me afterwards was a carbon-accounting one, not a grid one.

Most organisations calculate the emissions from the electricity they buy using a single annual average grid emission factor. Just one number, applied uniformly to every kWh, regardless of when it was consumed. With an engineering background and a working interest in carbon accounting, LCA, and data-backed decarbonisation, that annual-average assumption has always struck me as the weakest link in an otherwise rigorous accounting exercise. With some free time, curiosity got the better of me: how wrong, in practice, is a single annual number?

So I built the check myself, from public hourly grid data rather than from the published annual averages. The answer: quite wrong. An organisation's true electricity-related emissions can vary by an order of magnitude depending on when, not just how much it consumes power. The same underlying hourly data also shows why a naive national supply/demand indicator understates Germany's real network-driven renewable redispatch volume by roughly 17× - a separate but related measurement gap. Both findings share the same root cause: averaging over time (or over network location) destroys exactly the information a decarbonisation strategy needs.

## 2. Data and Method

The analysis uses 8,760 hourly observations for 2025: grid load and residual load, and generation by technology (biomass, hydropower, wind onshore/offshore, solar PV, other renewables, nuclear, lignite, hard coal, fossil gas, pumped storage, other conventional), all from SMARD (Bundesnetzagentur's market data platform). Monthly renewable redispatch-reduction volumes by technology and monthly congestion-management costs (redispatch, grid reserve, countertrading) are also from Bundesnetzagentur reporting.

Estimated operational carbon intensity is calculated hourly as combustion-based generation weighted by representative emission factors (lignite 1,020; hard coal 820; other conventional 700; fossil gas 490; biomass 200 gCO<sub>2</sub>/kWh), divided by total generation across all technologies. Biomass carries a non-zero operational factor here (reflecting direct combustion emissions) even though it remains classified as renewable generation for the purposes of the renewable-share metric, a deliberate departure from treating biomass as zero-emission at the point of combustion, and itself a small illustration of Finding 1: a single label ("renewable") can obscure more than one real quantity. This mirrors, at hourly resolution, the same logic that GHG Protocol Scope 2 location-based accounting applies at annual resolution: emissions attributed to a kWh based on the generation mix supplying the grid at the time it was consumed. The difference here is only the time step — hourly instead of annual — and that difference turns out to matter a great deal (Section 3).

### 3. Finding 1: The Annual Average Hides an 8.7× Range

Estimated hourly carbon intensity ranged from 76 to 665 gCO<sub>2</sub>/kWh across 2025 shows a factor of 8.7, against an annual average of 331 gCO<sub>2</sub>/kWh. That annual average is the number a standard location-based Scope 2 calculation would use. Figure 1 shows what it is quietly averaging over: a well-defined low-carbon window, roughly 10:00–16:00 between April and September, driven by high solar output, surrounded by a much larger high-carbon region covering winter evenings, nights and early mornings.

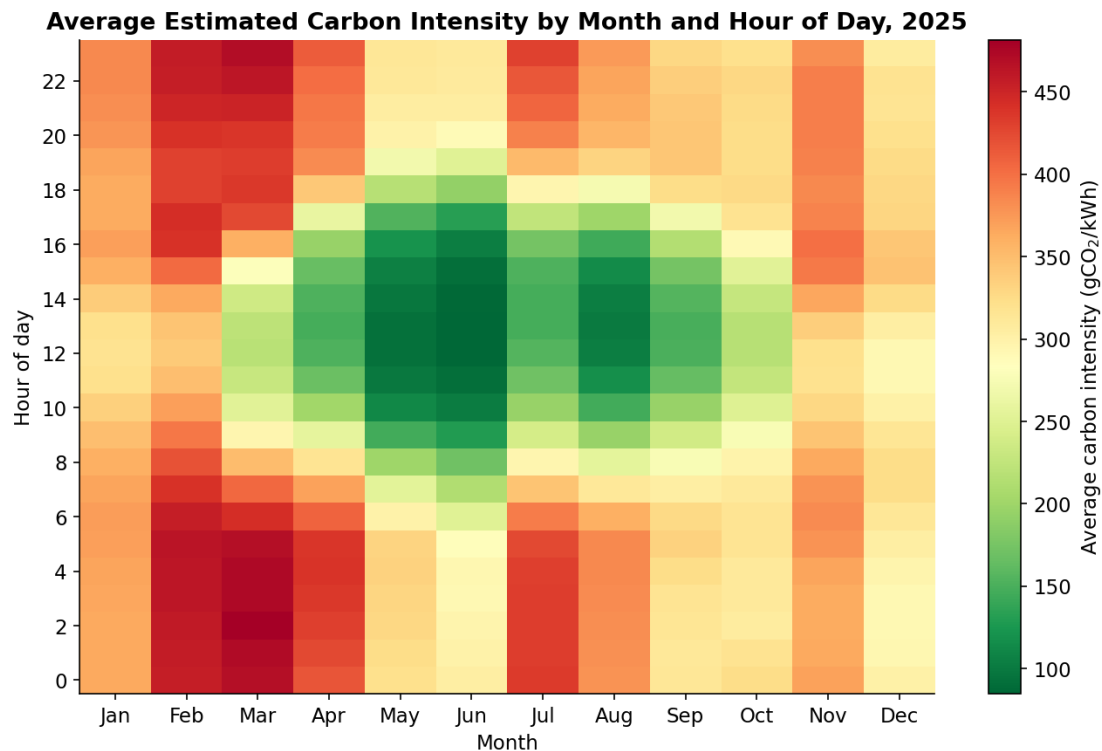


Figure 1. Average estimated carbon intensity by month and hour of day, 2025.

The practical consequence: two organisations with identical annual electricity consumption in Germany can have genuinely different carbon footprints. Hence, potentially differing by a large multiple, purely because of when their load occurs (a data centre running flat out overnight vs. a facility that shifts load into the solar-rich midday window), and a standard annual-average Scope 2 factor cannot see that difference at all. This is precisely the gap that hourly/24-7 carbon-free-energy accounting approaches (e.g. the EnergyTag standard, Google’s and Microsoft’s 24/7 CFE reporting) are designed to close, and this dataset shows why: demand itself is a poor proxy for carbon intensity ( $r = 0.025$  between hourly grid load and estimated carbon intensity), so a load-shifting strategy has to target the carbon signal directly rather than assume low demand means low carbon.

### 4. Finding 2: The Same Blind Spot at a System Level

The same averaging problem shows up again at system level. Negative residual load (hours where wind and solar generation alone exceeded national demand) occurred in 202 hours in 2025 (2.3% of the year), totalling 552 GWh of potential-surplus energy by magnitude. Mapped by month and hour (Figure 2), this sits almost exactly on the same low-carbon window as Figure 1, as expected: both are driven by the same wind and solar availability.

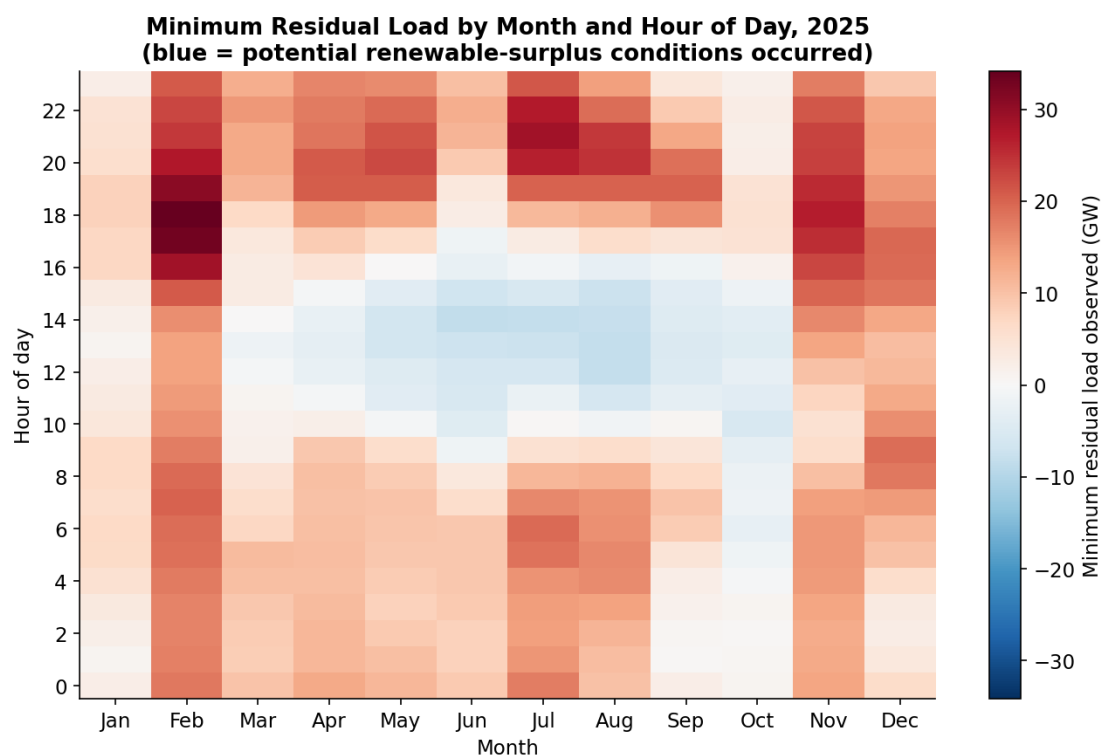


Figure 2. Minimum residual load by month and hour of day, 2025 (blue = surplus conditions occurred).

But 552 GWh is not the real scale of the problem. Actual renewable redispatch reductions, based on the TSOs' own record of renewable generation ordered offline for network reasons totalled 9.49 TWh in 2025, roughly 17 times larger, and close to the 9.4 TWh Germany curtailed in 2024. The gap ranged from about 4× to more than 250× by month. In addition, redispatch occurred in every month including four months with zero negative-residual-load hours at all.

| Metric                                | This analysis (2025) | Published reference          |
|---------------------------------------|----------------------|------------------------------|
| Total congestion-management cost      | €3,079m              | €3.07bn (2025, BNetzA)       |
| Renewable redispatch / curtailment    | 9.49 TWh             | 9.4 TWh (2024, BNetzA)       |
| National potential-surplus indicator  | 552 GWh              | n/a — not an official series |
| Surplus-indicator vs. real redispatch | ≈17× understatement  | —                            |

Table 1. This analysis's totals against independently published Bundesnetzagentur figures.

The close match to the published figures (Table 1) is a useful cross-check on the underlying data. The gap between the national indicator and the real redispatch volume is the more interesting result: it is the same failure mode as Finding 1, one level up. A single national supply/demand aggregate hides a geographically concentrated problem (wind in the north, demand and retiring conventional capacity in the south, transmission that has not kept pace) in exactly the way a single annual carbon factor hides a temporally concentrated one. Averaging over the wrong dimension: time, or geography makes a real problem statistically disappear.

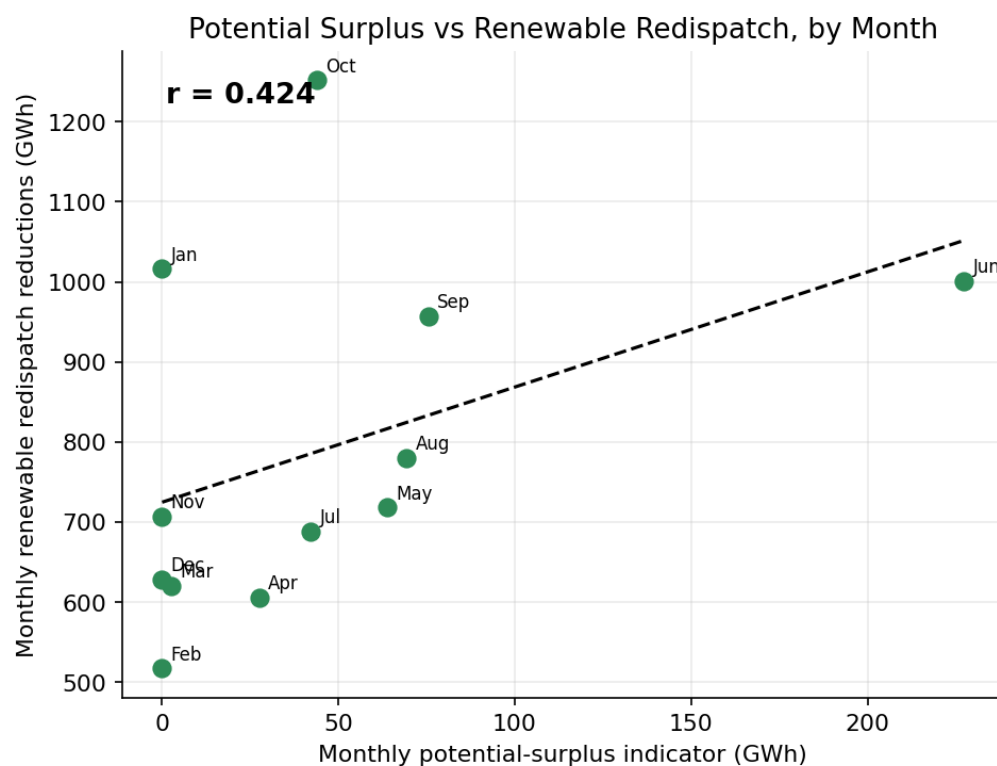


Figure 3. Monthly potential-surplus indicator vs. renewable redispatch reductions, 2025 ( $r \approx 0.42$ ).

## 5. What This Means for Carbon Accounting and Decarbonisation Analytics

Three implications follow directly, aimed at practice rather than policy:

- **Annual grid factors understate what good data could show.** Organisations with flexible load (EV fleets, heat pumps, batch industrial processes, data centres) could materially change their reported footprint by shifting consumption toward low-carbon hours. This is a lever that is invisible under annual-average Scope 2 accounting and only becomes visible at hourly resolution.
- **Location-based and market-based methods both need a granularity upgrade.** This is the same argument driving the shift toward hourly/24-7 carbon-free-energy matching in leading corporate reporting. It does not require new instrumentation as SMARD-equivalent public hourly grid data already exists in most European markets; it requires accounting practice to use it.
- **The pattern generalises beyond electricity.** Averaging over the wrong dimension such as time here, but equally geography, product line, or supplier tier in an LCA or Scope 3 context is a general way for a sustainability metric to be technically correct and substantively misleading. Checking a headline number against granular, independent data is a habit worth building into any decarbonisation analysis, not just this one.

## 6. Limitations

This analysis is deliberately transparent about what it does not show. Emission factors are representative published values, not plant-specific, and the carbon-intensity estimate is generation-based, excluding imports/exports and lifecycle (embodied) emissions as a full LCA treatment would need to add upstream fuel-cycle and infrastructure emissions on top of this operational estimate. Redispatch and cost data are monthly and technology-level only; hourly, network-level, or regional data would allow the ~17× gap in Section 4 to be attributed to specific constraints rather than inferred nationally. No electricity-price data was included, so the analysis cannot yet separate carbon-aware from price-aware load-shifting incentives.

## 7. Conclusion

A single annual number is convenient, auditable, and on this evidence, hiding an 8.7× range that matters for anyone trying to genuinely reduce, rather than just report, electricity-related emissions. The same blind spot, one level up, is most of why a naive national supply/demand indicator understates Germany's real network-driven renewable redispatch volume by roughly 17×. Both are the same lesson applied at different scales: good decarbonisation analysis has to go and check what the headline number is averaging over. This was just a few evenings with public data proving that point to myself; I'd be glad to compare notes with anyone working on the same problem from the accounting, LCA, or grid-analytics side.

Methodology, full hourly dataset, Excel workbook, and report available on github (link sent on request).

Data sources: SMARD (Bundesnetzagentur); Bundesnetzagentur redispatch and congestion-cost reporting.